# Production methodology: A4

Companion to **A Switching Dilemma for Normality in Actual Causation**, version 10, as adopted with one correction (patch P1). Prepared 30 October 2026; updated 31 October 2026.

The matching essay contains **5,825 words** under the conservative source count, including all non-bibliographic material. This anonymized report covers production, revision and verification; it carries forward the preceding production report and adds the readability revision described below.

## Production setting and models

The essays were developed and revised during 21–31 October 2026 through two AI systems working with a shared project repository. This report reconstructs the production method from the retained drafts, review reports, source comparisons and production summary. It describes recorded work rather than treating a planned workflow as a completed experiment. Some historical details remain unverified.

The Claude-side records identify the scaffold designer, orchestrator, drafting agents and review agents as **Claude Opus 5.5**, with model string `claude-opus-5-5`; the orchestrator's environment was recorded as Claude Code 2.1.306 initially and 2.1.314 later in the run. These identifiers are reported from that system's records, not independently recovered from service telemetry. The other system was an **OpenAI Codex assistant**. A verified session-by-session model-version record for its earlier work was not available. It contributed candidate generation, critical review, mathematical checks, source comparison, substantive revisions and final document preparation. The preceding substantive finishing pass and the current readability revision also used Codex. Reviews by a third model (Grok), relayed by the operator on 31 October, and the Claude-side final audit and adoption on 31 October are described in the section "Final audit and adoption" below.

## Candidate generation and selection

The initial AI-written scaffold proposed a broad funnel: survey the literature, generate candidates, reject anticipated or defective claims, draft survivors, obtain critical reviews, revise, audit citations and proofs, and package the results. Public research interests of the intended academic audience informed the literature areas and reviewer personas. The reviewer personas were composite subfield profiles given to models as prompts; none represented, imitated or involved any individual scholar. No scholar's participation or endorsement is claimed.

The original plan was reduced after budget and usage constraints. Three early essays were drafted from literature maps; prior-art checks led to replacement of some initial theses. A later problem survey listed 30 problems, shortlisted ten and selected six. Source availability helped determine which three were developed first. For these three, ideation agents produced fifteen proposals, five per topic. Recorded critical assessments rejected eight and repaired or merged seven into three essay plans. An additional Codex idea search generated 120 candidates and narrowed them through 40, 16 and eight; its survivors did not become these final essays. One other essay was withdrawn after its central objections remained unresolved.

Selection ultimately used the models' recommendations and the operator's choice among AI-produced outputs. The proposed formal draft tournament was not run. The process was exploratory and adaptive, not a preregistered evaluation. This matters because the final essays were selected from substantially more attempts than the three displayed here.

## Instructions, scaffolding and revision

The recurring instruction pattern was to reconstruct the target's strongest argument, identify its assumptions, compare the closest primary literature, propose a precise increment, look for counterexamples, repair errors and narrow unsupported conclusions. Ideation tasks asked for multiple candidate arguments and an explicit account of what each would add. Critical tasks asked models to find prior art, invalid steps, misleading attributions and objections that would survive a charitable reading. Drafting tasks required a self-contained philosophical argument within the word limit. These are descriptions of the prompt functions, not a verbatim prompt archive.

Claude's workflow assigned separate roles for literature surveying, ideation, critical testing, drafting, revision, citation checking, formal checking and document checks. Some roles existed only in the planned scaffold; an editor stage and the proposed multi-round ranking tournament were not completed. Selected drafts received anonymous model-review packets containing the manuscript, a rubric and a main source. Some packets contained an additional comparator. The review roles included a subject specialist and an advocate of the position criticized. Their reports informed substantive revisions. Separate model instances and anonymous packets reduce some contextual influence but do not create human expert review or establish statistical independence between judgments.

Codex reviewed successive drafts against primary sources and supplied counterexamples, calculations and restrictions on claims. The two systems then incorporated each other's revisions. The last Claude pass added citation and proof-audit corrections. The preceding substantive edition incorporated those corrections and reviewed the changed claims. The current edition improves exposition and produces matching manuscripts and PDFs. It does not count a proposed check as completed or describe a previous draft's review as a new review of this version.

## Research and computational tools

The systems used web search, author-posted papers, journal or repository copies, uploaded texts, text extraction and the shared repository. Bibliographic metadata checks supplemented reading; abstracts and metadata were not treated as substitutes for a full-text check of a theorem. Essential inaccessible texts were requested as documents rather than as human explanations.

Computational work used Python, exact rational arithmetic, symbolic algebra, finite-model enumeration, numerical integration and, in the Claude-side causation audit, an SMT solver. Such calculations checked examples and finite domains. Written proofs support claims quantified over all decision problems or all models in a stated class. Neither numerical agreement nor finite enumeration establishes originality. The essay-specific section below distinguishes checks performed during substantive review from the document checks repeated for this readability edition.

PDFs were generated from the final Markdown sources using Pandoc and XeLaTeX. The existing counting script counts title, abstract, headings, mathematical tokens, footnotes and appendices,

excluding only the bibliography. Text and metadata were checked for identifying strings, and rendered pages were visually inspected. The combined reading copy is an additional convenience document; the individual essay and its matching report are the paired documents.

## Human involvement and context

The operator set the task and deadline, requested more candidates and stronger originality and significance, supplied or relayed source documents and model reviews, asked for further revision, and selected among AI-generated work. Recorded methodological suggestions included using relevant philosophical perspectives and demanding criticism. The available project review identified no substantive philosophical argument, example, distinction or repair supplied by the operator that entered these essays. This is a limited provenance finding, not a certification of every earlier session.

Claude's records state that automatic memory was disabled and personal-account retrieval was not used in its production workflow. A previous Codex disclosure recorded possible availability of unrelated personal context; the subsequent limited audit identified no human philosophical contribution but did not certify all historical context. That limitation remains disclosed. The preceding substantive pass did not retrieve personal context or use unrelated task histories to develop arguments. Raw conversations, human-input histories and internal run logs are not included in this report.

## Resources and limitations of the record

The Claude-side usage tool, rerun on 31 October 2026 at 13:29 UTC, reports 3,571 requests to Claude Opus 5.5, including 3,173 subagent requests, across 86 transcript files covering 21 October 18:25 UTC to 31 October 13:29 UTC, and estimates approximately US$476.44 at list prices. The transcripts show activity in 72 ten-minute intervals, approximately twelve hours of occupied intervals across that period; this is not a measurement of processor time or summed agent time. These final figures cover the whole Claude-side run for all three essays, including the final audits, but not the last reconciliation of these reports. They are list-price estimates, not a bill: work used subscription access with usage limits. Reliable cumulative Codex token use, cost and machine time were unavailable. Total human time and a defensible cost allocation to this particular essay could not be established.

The earlier anonymous model reviews were critical; no calibrated human-expert assessment of these final versions has been completed. The workflow cannot demonstrate that the literature search was exhaustive, that model judgments are independent, or that the surviving contributions are important enough for publication. The most consequential remaining questions are stated below. No contest submission has been made as part of the preceding substantive pass.

## Essay-specific development and checks

The initial proposals concerned the switch–preemption contrast, calibration of normality and a frozen-witness lemma. A critical test supplied the Doorbells model and coding comparisons. Later Codex revisions repaired the lemma, restricted the scope and compared the argument with book-length treatments. A model referee's cross-definition objection led to the blocked-tracks case

and the current pair of witness-profile theorems. Further review restricted the additive escape argument and made the fixed-context assumptions explicit.

Halpern's *Actual Causality* was available as an uploaded full book. The prior source comparison inspected the relevant normality and preemption discussion, including pages 38–39, 79–81 and 90–91, and the conservative-extension reasoning used in the lemma comparison. Wysocki's *Conjoined Cases* was read as a close comparator: it gives a stronger obstruction in a different setting. The present essay's proposed increment is the cross-model profile identity and cross-definition comparison, not a general impossibility of normality-based causation. The preceding substantive pass also checked Gallow's Doorbells discussion on pages 85–86 and the final conditions in Andreas and Günther's 2024 regularity theory against primary texts.

Earlier checks inspected the witness arguments and evaluated the boundary additive score $(r, a, b, c, d) = (2, 0, -1, -5, -3)$: the blocked-tracks witness has gain 3 and the clear-tracks witness gain 1. Claude's subsequent formal audit records a separate implementation of the three definitions, enumeration of 355 preorders, exact SMT checks, and additional tests of the conformity lemma. Those complete programs were not rerun for this edition. Their reports are computational support; the written witness arguments carry the impossibility claims. The regularity-theory calculations in Section 7 remain author-computed rather than independently reimplemented here.

The last Claude corrections replace "total orders" with "total preorders," remove an overinclusive claim about Billy's throw, repair the account of additional switch contingencies, specify the variable-indexed escape, and correct the account of Gallow's model, which retains the bells. This edition preserves those corrections. The mathematical content of this essay is unchanged from Claude's final manuscript at the reviewed checkpoint.

The main remaining assessment is philosophical significance relative to the existing literature. The result is conditional on event coding, profile anonymity and the stated Pareto condition. Different model choices or principled non-anonymous information can escape it. The application to Xiao and Xiong remains conditional on availability being evaluated at a fixed context with nonintervened roots unchanged. A4 is therefore retained as the less secure third candidate rather than presented as equally strong evidence of a novel contribution.

## Readability revision

The revision explains the backup-route comparison, witness profiles and the difference between default values and equation conformity. It separates the three witness calculations and the repair discussion into readable stages and clarifies that conformity is assessed against the original non-root equations. The formal statements, equations and proof text are unchanged; proof paragraphs were divided without changing their wording. Manuscript comparisons confirmed preservation of the formal content, and word counts, anonymity checks and PDF layout checks were repeated. The final review described next followed this prose revision; earlier substantive checks are identified above.

## Final review

Codex rechecked the witness-profile and fixed-context arguments and the relevant Halpern book passages on late preemption and switches. The final Claude checkpoint made no further substantive change to this essay. The existing comparison with Wysocki was retained from the earlier source review; the final pass did not obtain a fresh full-text copy. The regularity-theory computations were not rerun. The remaining concern is the significance of the restricted profile result, not a newly demonstrated proof defect. This was further review by an AI system involved in developing the essays, not independent expert endorsement.

## Submission pass

The submission pass re-read the complete manuscript and checked the witness-profile arguments and frozen-witness lemma. It added an explicit acknowledgment that Sprinkler applies an earlier concern of Halpern's to a comparison with switching. The page reference first given for that concern could not be verified in the final audit and was replaced (see below); the adopted text reads: "Halpern (2016, 84–85) already notes that Suzy's throw stops being a cause by itself when Billy's throwing without hitting is too abnormal." The theorem statements, witness calculations, proofs and numerical claims are unchanged. The remaining reservation concerns the philosophical significance of the restricted policy class; the earlier computation suites were not rerun.

## Final audit and adoption (31 October 2026)

Before adoption, a Claude-side citation, formal and compliance delta audit compared version 10 with the previously audited version 8 (5,581 words). It found thirteen changed passages: nine new or rewritten prose passages, one new attribution with a page number, and paragraph splits. The references are byte-identical to version 8 and the quotations identical; all 193 mathematical spans and every theorem, lemma and corollary statement are identical; the Theorem 1 proof is only divided into paragraphs. The Claude-side independent implementation of the three definitions was rerun, and its output was byte-identical to the stored output (94 checks: 91 pass, and the 3 failures are designed negative controls).

The new attribution said that Halpern illustrates the problem of a reliable backup with an automatic rock-throwing machine on page 85. No retrieved copy supported it, and the recorded full-book review, which covers chapter 3, does not mention such a machine; it supports pages 84–85 for a different point. Under the rule against unverified attribution, the Claude orchestrator applied patch P1 verbatim, replacing that sentence with: "Halpern (2016, 84–85) already notes that Suzy's throw stops being a cause by itself when Billy's throwing without hitting is too abnormal." The following sentence, "Sprinkler applies that concern to a comparison with switching," is unchanged. The pages 84–85 point was already cited in version 8; its confirmation rests on the recorded full-book review, not on pages retrieved on the Claude side. A request for the printed page 85 was filed and had not been fulfilled at adoption. With P1 the essay has 5,813 words. Injection and anonymity checks showed no hard flags.

On 31 October the operator relayed reviews of all three essays by a third model (Grok), saved verbatim. The Claude-side record of them notes that they reported no false lemma, no misrepresented source passage and no prior work already establishing the claimed contribution; their requests

concern framing and exposition. The review of this essay recommended major revisions. Version 10 predates it. The delta audit mapped each request against version 10: the request to defend the significance of the anonymity baseline is partly addressed, mostly by text already present in version 8; the requests concerning parts of causes, independently motivated event coding, explicit uniqueness arguments for the modified definition, a table of categories and an anonymity-like restriction for the regularity theory are not newly addressed; the Wysocki increment, the gloss on the additive identity and the admission about model aptness were already present. No change was made in response to the review beyond what version 10 already contained.

## Final document pairing

The adopted essay is version 10 (`A4-Causation-v10.md`) with patch P1 applied; the anonymous essay PDF was rebuilt from the patched text. This report is paired with that text.

## Late review edit (31 Oct)

An outside AI model (DeepSeek) wrote referee reports on the final essays, relayed unedited by the operator. A Claude-side substance-only triage found no false theorem but two wording inconsistencies, both corrected: a sentence that used "componentwise" for a variable-indexed escape that the essay's definition excludes, and a compensation claim now restricted to the modified definition. It also added "by itself" to make the single-candidate reading explicit. No theorem statement or computation changed. Final length: 5,825 words.